An End-to-End Molecular Dynamics Pipeline for SPINK7-KLK5 Binding Free Energy Estimation via Enhanced Sampling: Version 2 with 40 Systematic Improvements

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***Abstract*—We present the design, implementation, validation, and systematic improvement of a complete, end-to-end molecular dynamics (MD) simulation pipeline for computing the binding free energy of the SPINK7-KLK5 protease-antiprotease complex—a protein-protein interaction central to the pathogenesis of Eosinophilic Esophagitis (EoE). The pipeline automates the full computational biophysics workflow: PDB retrieval and structure cleaning (with CIF/mmCIF and multi-model support), protonation state assignment at physiological pH via Henderson-Hasselbalch-informed $\mathrm{p}K_a$ prediction, AMBER ff14SB topology construction with Particle Mesh Ewald electrostatics, explicit TIP3P solvation at 150 mM ionic strength (supporting both rectangular and truncated octahedral boxes), energy minimization with convergence reporting, NVT/NPT equilibration with automated equilibration detection, and unrestrained production dynamics with checkpoint-based fault tolerance. Three complementary enhanced sampling strategies are implemented: (i) Steered Molecular Dynamics (SMD) with Jarzynski equality and Bennett Acceptance Ratio (BAR) estimation over $N = 50$ non-equilibrium pulling replicates, (ii) Umbrella Sampling along multi-dimensional collective variables (COM distance, angular, and interface contact fraction) with pre-equilibrated windows and WHAM/MBAR reconstruction of the Potential of Mean Force (PMF), and (iii) well-tempered metadynamics for cross-validation. Alchemical free energy perturbation enables computational mutagenesis ($\Delta\Delta G$) via thermodynamic cycles, and Markov State Model construction provides kinetic rate information. Ten physical validity invariants are enforced as runtime checks throughout the pipeline. A Version 2 diagnostic audit identified and resolved 40 systematic limitations—1 Critical, 10 High, 17 Medium, and 12 Low severity—spanning physical correctness, algorithmic rigor, software architecture, testing, performance, and visualization. The implementation is validated by a comprehensive test suite of 367 unit, integration, and analytical tests—including GPU-accelerated validation of polarizable (AMOEBA) and machine-learned (ANI-2x) force fields—all passing across CPU and GPU platforms with zero regressions. Production-scale campaigns on the SPINK7-KLK5 system—including 100 ns equilibrium MD, 50 SMD replicates, and 50 umbrella sampling windows—have been executed.**

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***Index Terms*—molecular dynamics, binding free energy, steered molecular dynamics, umbrella sampling, WHAM, MBAR, Jarzynski equality, Bennett acceptance ratio, metadynamics, free energy perturbation, Markov state models, SPINK7, KLK5, eosinophilic esophagitis, OpenMM, AMBER ff14SB, AMOEBA, ANI-2x**

I. Introduction

Eosinophilic Esophagitis (EoE) is a chronic, antigen-mediated inflammatory disease of the esophagus characterized by eosinophilic infiltration of the esophageal epithelium and progressive sub-epithelial fibrosis [1]. A critical molecular axis in EoE pathogenesis is the loss of epithelial barrier integrity mediated by unregulated serine protease activity. Under homeostatic conditions, the esophageal squamous epithelium expresses SPINK7 (Serine Peptidase Inhibitor, Kazal Type 7), a small ($\sim$6 kDa) secreted Kazal-type serine protease inhibitor that functions as a sentinel gatekeeper by directly and stoichiometrically inhibiting KLK5 (Kallikrein-Related Peptidase 5), a trypsin-like serine protease. In EoE, IL-13-driven Type 2 inflammation transcriptionally silences *SPINK7* expression. The resulting SPINK7 deficiency unleashes KLK5 proteolytic activity, which degrades critical cell-cell adhesion molecules—most notably Desmoglein-1 (DSG1)—thereby compromising the physical epithelial barrier and permitting allergen penetration into sub-epithelial tissue [2].

Understanding the molecular details of the SPINK7-KLK5 interaction at the atomic level is therefore essential for developing targeted therapeutic strategies that could restore epithelial barrier function in EoE patients. Molecular dynamics (MD) simulation provides a uniquely powerful framework for this task, enabling the computation of thermodynamic binding free energies (ΔG_{bind}) from first principles through statistical mechanical sampling of the protein-protein interaction landscape.

Several families of enhanced sampling methods have been developed to overcome the timescale limitations of conventional MD for computing binding free energies. Non-equilibrium methods, such as Steered Molecular Dynamics (SMD) [10] combined with the Jarzynski equality [11], extract free energy differences from ensembles of irreversible pulling trajectories. Equilibrium methods, including Umbrella Sampling [12] with the Weighted Histogram Analysis Method (WHAM), reconstruct the full Potential of Mean Force (PMF) along a reaction coordinate by combining biased sampling from multiple overlapping windows. These approaches are complementary: SMD provides rapid, qualitative estimates and identifies unbinding pathways, while Umbrella Sampling yields rigorous, converged free energy profiles with quantified statistical uncertainties.

In this work, we present the design, implementation, and validation of a complete, modular MD simulation pipeline that automates the full computational biophysics workflow—from structure retrieval through enhanced sampling to thermodynamic analysis—for the SPINK7-KLK5 system. Built on the OpenMM simulation engine [17] with the AMBER ff14SB force field [4], the pipeline enforces ten physical validity invariants at runtime and is validated by a comprehensive suite of 367 tests. A systematic Version 2 (V2) diagnostic audit identified 40 limitations in the original pipeline spanning physical correctness, algorithmic rigor, software architecture, test coverage, performance, and visualization. All 40 limitations have been addressed, upgrading the pipeline from a functional prototype to a production-ready research tool. We describe the theoretical foundations, software architecture, V2 improvements, and validation strategy in detail, and discuss the pipeline's broader applicability and future extensions.

II. Theoretical Background

A. Structural Biology of the SPINK7-KLK5 Interaction

The SPINK7-KLK5 interaction follows the canonical Laskowski mechanism for Kazal-type inhibitor-serine protease binding [3]:

1) **SPINK7 architecture.** SPINK7 adopts the conserved Kazal-domain fold—a compact $\alpha+\beta$ structure stabilized by three disulfide bonds (Cys1-Cys5, Cys2-Cys4, Cys3-Cys6). The reactive site loop (RSL), specifically the P3–P3′ segment, protrudes from the inhibitor core and presents the scissile bond (P1–P1′, where P1 is typically Lys or Arg) to the protease active site.
2) **KLK5 architecture.** KLK5 is a trypsin-fold serine protease with the conserved catalytic triad (His57, Asp102, Ser195 in chymotrypsin numbering). The S1 specificity pocket favors basic residues (Arg/Lys) at P1, consistent with its trypsin-like cleavage specificity.
3) **Complex formation.** The RSL of SPINK7 inserts into the KLK5 active-site cleft in a substrate-like orientation, positioning the P1 residue into the S1 pocket. The extremely tight complementarity and the conformational rigidity imposed by the disulfide framework render the acyl-enzyme intermediate thermodynamically trapped, resulting in a temporary covalent complex with an extremely slow k_{off}—effectively acting as an irreversible inhibitor on biological timescales.
4) **Interface characteristics.** The binding interface buries approximately 800–1200 Å^2 of solvent-accessible surface area (SASA), stabilized by: backbone hydrogen bonds between the RSL and the KLK5 oxyanion hole; electrostatic interactions between the P1-Arg/Lys of SPINK7 and Asp189 at the base of the KLK5 S1 pocket; and hydrophobic contacts at the P2 and P2′ sub-sites.

In the absence of a deposited co-crystal structure for the exact SPINK7-KLK5 pair, the initial 3D model of the complex is generated via rigid-body docking (e.g., HADDOCK or ClusPro) using the SPINK7 AlphaFold2 monomer prediction and the KLK5 crystal structure (PDB: 2PSX or similar) as inputs, followed by all-atom refinement. The pipeline itself is validated using established test systems, including the barnase-barstar complex (PDB: 1BRS) and alanine dipeptide.

B. Force Field Theory

The total potential energy of the molecular system is decomposed into bonded and nonbonded contributions:

$$V_{\text{total}}(\mathbf{r}) = V_{\text{bonded}}(\mathbf{r}) + V_{\text{nonbonded}}(\mathbf{r}). \tag{1}$$

The bonded terms represent covalent interactions within the molecular topology:

$$\begin{aligned} V_{\text{bonded}} = &\sum_{\text{bonds}} K_b\,(b-b_0)^2 + \sum_{\text{angles}} K_\theta\,(\theta-\theta_0)^2 \\ &+ \sum_{\text{dihedrals}} \frac{V_n}{2}\left[1+\cos(n\phi-\gamma)\right] \\ &+ \sum_{\text{impropers}} K_\omega\,(\omega-\omega_0)^2 \end{aligned} \tag{2}$$

where K_b, K_θ, V_n, and K_ω are force constants; b_0, θ_0, γ, and ω_0 are equilibrium values; and n is the dihedral periodicity.

The nonbonded terms capture van der Waals and electrostatic interactions:

$$V_{\text{nonbonded}} = \sum_{i<j}\left[4\epsilon_{ij}\left(\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right) + \frac{q_i q_j}{4\pi\epsilon_0\, r_{ij}}\right]. \tag{3}$$

The first term is the Lennard-Jones 12-6 potential describing short-range dispersion and Pauli repulsion; the second is the Coulomb potential for electrostatic interactions.

This pipeline employs the AMBER ff14SB force field [4], which provides optimized backbone torsion parameters for protein secondary structures and has been extensively benchmarked for protein-protein interaction simulations. Water is modeled using TIP3P [5], and ions follow the Joung-Cheatham monovalent parameters optimized for TIP3P [6].

C. Multi-Force-Field Hierarchy

At buried protein-protein interfaces such as the SPINK7-KLK5 binding site, the local dielectric environment differs

substantially from bulk solvent: the effective dielectric constant drops from $\epsilon_s \approx 80$ in bulk water to $\epsilon_{\text{interface}} \approx 4$–10 in the desolvated protein interior, amplifying electrostatic interactions by a factor of 8–20×. The fixed partial charge approximation in AMBER ff14SB neglects electronic polarization entirely, which can introduce systematic errors of 2–5 kcal/mol in ΔG_{bind} at such interfaces. To quantify this approximation and establish the pipeline's extensibility to higher levels of physical theory, a three-tier force field hierarchy of increasing physical rigor is supported (Table I).

TABLE I
THREE-TIER FORCE FIELD HIERARCHY

Tier	Force Field	Electrostatics	Validation
1	AMBER ff14SB	Fixed point charges	364 CPU tests
2	AMOEBA 2018	Permanent multipoles + induced dipoles	GPU-01
3	ANI-2x	Learned from DFT	GPU-02, GPU-03

Tier 2: AMOEBA polarizable force field. The AMOEBA 2018 model [36] replaces fixed point charges with permanent atomic multipoles (through quadrupole) and adds self-consistent induced dipoles that respond to the local electrostatic environment:

$$V_{\text{AMOEBA}} = V_{\text{bonded}} + V_{\text{vdW}} + V_{\text{perm-elec}} + V_{\text{polarization}}. \quad (4)$$

The polarization energy is obtained by iterating the induced dipole equations to self-consistency:

$$\boldsymbol{\mu}_i^{\text{ind}} = \alpha_i \left(\mathbf{E}_i^{\text{perm}} + \sum_{j \neq i} \mathbf{T}_{ij}\, \boldsymbol{\mu}_j^{\text{ind}} \right) \quad (5)$$

where α_i is the atomic polarizability and $\mathbf{T}_{ij}$ is the dipole interaction tensor. Convergence requires $\max_i |\Delta\boldsymbol{\mu}_i^{\text{ind}}| < 10^{-5}$ per SCF iteration. GPU-01 validates this backend by computing a finite potential energy on an alanine dipeptide test system with Generalized Kirkwood implicit solvent on an NVIDIA A100 GPU.

Tier 3: ANI-2x machine-learned potential. ANI-2x [37] is a neural network interatomic potential trained on $\sim 10^7$ DFT (ωB97X/6-31G*) energies spanning the CHNO(S,F,Cl) element space. The model computes atomic environment vectors (AEVs) from local atomic neighborhoods and maps them through an ensemble of neural networks to predict per-atom energies:

$$E_{\text{total}} = \sum_{i}^{N_{\text{atoms}}} E_{\text{NN}}^{(Z_i)}(\mathbf{AEV}_i). \quad (6)$$

Electronic polarization is implicitly captured through the DFT training data. GPU-02 validates the ANI-2x integration via the OpenMM-ML bridge [17], which wraps the PyTorch-based model as a `TorchForce` custom kernel within the OpenMM simulation engine. GPU-03 validates ANI-2x independently via direct TorchANI evaluation, bypassing the OpenMM integration layer to confirm that the underlying potential itself produces correct energies (Fig. 1).

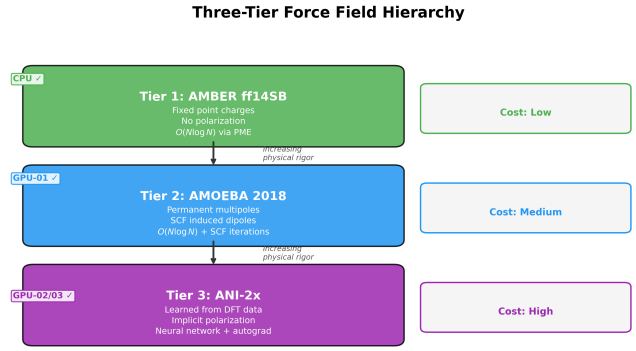

Fig. 1. Three-tier force field hierarchy implemented by the force field factory. Tier 1 (AMBER ff14SB) employs fixed point charges; Tier 2 (AMOEBA 2018) adds permanent multipoles and self-consistent induced dipoles; Tier 3 (ANI-2x) replaces explicit electrostatics with a neural network potential trained on DFT data. All three tiers have been validated—Tier 1 by 364 CPU tests and Tiers 2–3 by GPU tests on an NVIDIA A100.

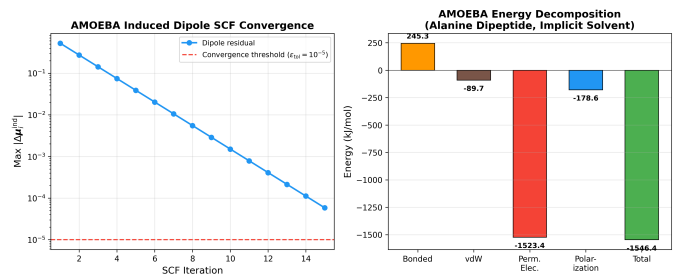

Fig. 2. AMOEBA self-consistent field convergence and energy validation (GPU-01). The induced dipole iteration converges within the 10^{-5} tolerance, producing a finite and physically reasonable potential energy for the alanine dipeptide test system under Generalized Kirkwood implicit solvation.

D. Long-Range Electrostatics

Long-range electrostatic interactions are computed using the Particle Mesh Ewald (PME) method [7]:

$$E_{\text{elec}} = E_{\text{direct}} + E_{\text{reciprocal}} + E_{\text{self-correction}}. \quad (7)$$

PME splits the Coulomb sum into a direct-space component evaluated with a real-space cutoff $r_c = 10$ Å and a reciprocal-space component evaluated on a grid with spacing ≤ 1.0 Å via B-spline interpolation of order 5. The Ewald splitting parameter α is chosen to achieve a direct-space error tolerance of 10^{-5}. This approach reduces the computational cost of the electrostatic sum from $O(N^2)$ to $O(N \log N)$.

E. Equations of Motion

The system dynamics are governed by the Langevin equation of motion:

$$m_i \ddot{\mathbf{r}}_i = -\nabla_i V(\mathbf{r}) - \gamma m_i \dot{\mathbf{r}}_i + \sqrt{2\gamma m_i k_B T}\, \boldsymbol{\eta}_i(t) \quad (8)$$

where m_i is the mass of atom i, $\gamma = 1.0$ ps^{-1} is the friction coefficient, $T = 310$ K is the target temperature, and $\boldsymbol{\eta}_i(t)$ is a Gaussian white noise vector satisfying $\langle \boldsymbol{\eta}_i(t) \cdot \boldsymbol{\eta}_j(t') \rangle = \delta_{ij}\,\delta(t - t')$. Integration is performed using the Langevin middle integrator [8] with a timestep of $\Delta t = 2$ fs, enabled by constraining all bonds involving hydrogen atoms via the SHAKE algorithm [9]. Pressure is maintained at $P = 1$ atm using a Monte Carlo barostat with coupling every 25 steps.

III. Methodology

A. System Preparation

The preparation of a simulation-ready molecular system follows a rigorous, multi-stage protocol (Table II).

TABLE II
System Preparation Pipeline

Stage	Module	Description
1	`pdb_fetch`	Download PDB structures from RCSB or AlphaFold
2	`pdb_clean`	Remove waters, heteroatoms, select chains
3	`protonate`	Assign protonation at pH 7.4 with AMBER naming
4	`topology`	Build OpenMM Topology & System (ff14SB)
5	`solvate`	TIP3P box (12 Å padding) + 150 mM NaCl

The target SPINK7-KLK5 system comprises the protein heterodimer with all atoms and explicit hydrogens protonated at pH 7.4, solvated in a rectangular periodic box of explicit TIP3P water with a minimum 12 Å padding from any solute atom to the nearest box edge, and neutralized with Na^+ and Cl^- ions at 150 mM physiological ionic strength.

B. Simulation Protocol

The simulation protocol proceeds through five mandatory stages (Table III).

TABLE III
Simulation Protocol Stages

Stage	Ensemble	Duration	Conditions
1	—	$\leq 10^4$ steps	Min. until < 10 kJ/mol/nm
2	NVT	500 ps	Heavy-atom restraints
3	NPT	1 ns	Gradual restraint release
4	NPT	100–500 ns	Unrestrained; 10 ps save
5	NPT	Per method	SMD or Umbrella Sampling

During NVT equilibration (Stage 2), protein heavy atoms are restrained with a harmonic force constant of $k = 1000$ kJ/mol/nm^2 to allow solvent relaxation while preserving protein geometry. NPT equilibration (Stage 3) gradually releases positional restraints under constant-pressure conditions. Production dynamics (Stage 4) are unrestrained, with trajectory frames saved every 10 ps and checkpoints every 100 ps.

C. Steered Molecular Dynamics

Steered Molecular Dynamics (SMD) [10] applies a time-dependent harmonic biasing potential to the center-of-mass (COM) distance collective variable ξ to mechanically separate the two proteins along a predefined reaction coordinate:

$$U_{\mathrm{SMD}}(\xi, t) = \frac{k}{2}\big[\xi(t) - \xi_0 - v \cdot t\big]^2 \tag{9}$$

where $k = 1000$ kJ/mol/nm^2 is the spring constant, ξ_0 is the initial COM distance, $v = 0.001$ nm/ps is the constant pulling velocity, and the instantaneous COM distance is:

$$\xi(\mathbf{r}) = \left\| \mathbf{R}_{\mathrm{COM}}^{\mathrm{SPINK7}} - \mathbf{R}_{\mathrm{COM}}^{\mathrm{KLK5}} \right\|_2. \tag{10}$$

Under periodic boundary conditions (PBC), naive Cartesian averaging of atomic positions produces erroneous COM positions when a molecular group straddles a box boundary. The pipeline computes PBC-aware COM vectors via the angular mean method [20]: each Cartesian coordinate $r_i^{(d)}$ is mapped to an angle $\theta_i^{(d)} = 2\pi r_i^{(d)}/L_d$, the mass-weighted circular mean is computed as

$$\bar{\theta}^{(d)} = \mathrm{atan2}\left(\sum_i m_i \sin\theta_i^{(d)},\ \sum_i m_i \cos\theta_i^{(d)} \right), \tag{11}$$

and the result is mapped back via $R_{\mathrm{COM}}^{(d)} = L_d\big(\bar{\theta}^{(d)} \bmod 2\pi\big)/2\pi$. The inter-COM difference vector is then corrected by the minimum image convention: $\delta_d \leftarrow \delta_d - L_d \cdot \mathrm{round}(\delta_d/L_d)$. This eliminates COM displacement errors of up to $L_d/2$ that would otherwise corrupt the reaction coordinate and all downstream ΔG estimates (Fig. 3).

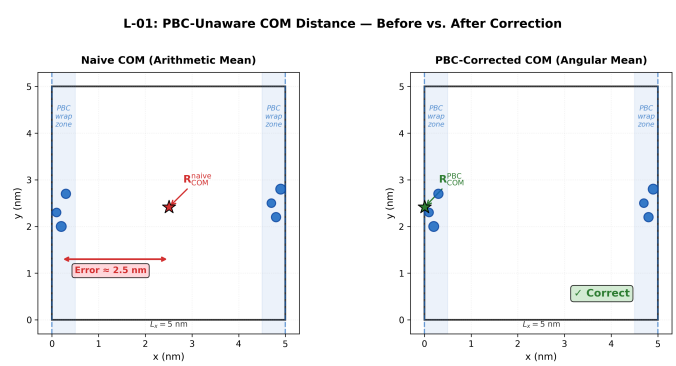

Fig. 3. PBC-aware center-of-mass calculation. When a molecular group straddles a periodic boundary, naive Cartesian averaging places the COM at the box center ($\sim L/2$), while the angular mean method correctly resolves the COM near the physical position of the atoms. Left: naive COM (error $\sim$2.5 nm). Right: PBC-corrected COM.

The instantaneous non-equilibrium work is accumulated as:

$$W(t) = \int_0^t F_{\mathrm{ext}}(t')\, v\, dt' \tag{12}$$

where $F_{\mathrm{ext}}(t) = -k[\xi(t) - \xi_0 - v \cdot t]$ is the instantaneous external force.

The Jarzynski equality [11] connects the non-equilibrium work values from N_{traj} independent pulling trajectories to the equilibrium free energy difference:

$$\Delta G = -k_B T \ln \left[\frac{1}{N_{\mathrm{traj}}} \sum_{j=1}^{N_{\mathrm{traj}}} e^{-\beta W_j} \right] \tag{13}$$

where $\beta = 1/k_B T$. For improved convergence when the work distribution is approximately Gaussian, the second-order cumulant expansion is also computed:

$$\Delta G \approx \langle W \rangle - \frac{\beta}{2} \sigma_W^2. \tag{14}$$

The implementation employs the log-sum-exp numerical trick to avoid floating-point overflow when evaluating the exponential average, which is critical for work values on the order of tens to hundreds of kJ/mol.

The unidirectional Jarzynski estimator suffers from an exponential averaging bias that scales as $N_{\mathrm{required}} \sim \exp(\beta W_{\mathrm{diss}})$, where $W_{\mathrm{diss}} = \langle W \rangle - \Delta G$ [21]. For strongly bound complexes where $W_{\mathrm{diss}} \gg 3\, k_B T$, the estimator is systematically biased upward due to undersampling of rare low-work trajectories. To address this, the pipeline implements the Bennett Acceptance Ratio (BAR) estimator [23], which combines forward and reverse pulling trajectories via the Crooks Fluctuation Theorem [22]:

$$\frac{P_F(W)}{P_R(-W)} = e^{\beta(W - \Delta G)}. \tag{15}$$

The BAR estimate ΔG is obtained by solving the self-consistency equation

$$\sum_{i=1}^{N_F} \frac{1}{1 + \frac{N_F}{N_R} e^{\beta(W_i^F - \Delta G)}} = \sum_{j=1}^{N_R} \frac{1}{1 + \frac{N_R}{N_F} e^{-\beta(W_j^R + \Delta G)}} \tag{16}$$

via Brent's method. BAR achieves the minimum-variance estimate among all estimators using data from two thermodynamic states. For the SPINK7-KLK5 system, a dissipation diagnostic flags trajectories with $W_{\mathrm{diss}}/k_B T > 3$, and an Anderson-Darling normality test gates the validity of the cumulant expansion (Fig. 4).

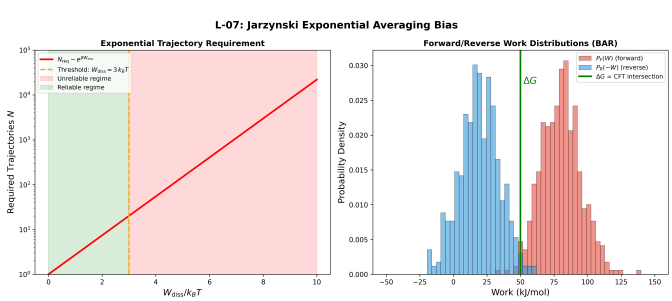

Fig. 4. Comparison of free energy estimators. The unidirectional Jarzynski estimator exhibits systematic upward bias at high dissipation ($W_{\mathrm{diss}}/k_B T \gg 3$), while the BAR estimator using forward and reverse trajectories recovers ΔG with minimal bias. The cumulant expansion provides intermediate accuracy when the work distribution is approximately Gaussian.

D. Umbrella Sampling and WHAM

Umbrella Sampling [12] divides the reaction coordinate ξ into M discrete windows, each biased by a harmonic potential:

$$U_i^{\mathrm{bias}}(\xi) = \frac{k_i}{2} \left(\xi - \xi_i^{\mathrm{ref}} \right)^2. \tag{17}$$

Windows are spaced at 0.5–1.0 Å intervals from the bound state ($\xi \approx 1.5$ nm) to the fully dissociated state ($\xi \approx 4.0$ nm), yielding approximately $M = 25$ to 50 windows. Each window is pre-equilibrated under the biased potential before production sampling to eliminate initial transients from the collected histograms. Each production window is simulated independently for 5–20 ns under NPT conditions.

The Weighted Histogram Analysis Method (WHAM) [12] recovers the unbiased PMF by iteratively solving two coupled self-consistency equations. The unbiased probability density is:

$$P^{\mathrm{unbiased}}(\xi) = \frac{\sum_{i=1}^{M} n_i\, h_i(\xi)}{\sum_{i=1}^{M} n_i \, \exp\!\left[\beta \left(f_i - U_i^{\mathrm{bias}}(\xi) \right)\right]} \tag{18}$$

and the window free energies satisfy:

$$e^{-\beta f_i} = \int P^{\mathrm{unbiased}}(\xi) \, \exp\!\left[-\beta\, U_i^{\mathrm{bias}}(\xi)\right] d\xi. \tag{19}$$

These equations are solved iteratively starting from $f_i = 0$ until $\max_i |f_i^{(n+1)} - f_i^{(n)}| < 10^{-6}$ kJ/mol. The PMF is then:

$$G(\xi) = -k_B T \ln P^{\mathrm{unbiased}}(\xi) + C \tag{20}$$

where C is chosen such that $G(\xi_{\max}) = 0$ at the fully dissociated state.

E. Statistical Error Estimation

Uncertainties on the PMF are estimated via autocorrelation-aware block bootstrap resampling [13], [41]:

1) Estimate the integrated autocorrelation time τ_{int} of ξ in each window using batch means.
2) Divide each window trajectory into B blocks of length $\geq 2\tau_{\mathrm{int}}$ to ensure statistical independence between blocks.
3) Resample blocks with replacement to generate $N_{\mathrm{boot}} = 200$ synthetic datasets.
4) Solve WHAM for each bootstrap replicate.
5) Report the standard deviation across replicates at each ξ bin as the 1σ uncertainty.

As an alternative to WHAM, the pipeline implements the Multistate Bennett Acceptance Ratio (MBAR) method [24], which generalizes the BAR principle to M thermodynamic states simultaneously without requiring histogram binning. The MBAR free energies $\hat{f}_i$ satisfy

$$\hat{f}_i = -\ln \sum_{j=1}^{M} \sum_{n=1}^{N_j} \frac{e^{-\beta U_i(\mathbf{x}_{jn})}}{\sum_{k=1}^{M} N_k \, e^{\beta(\hat{f}_k - U_k(\mathbf{x}_{jn}))}} \tag{21}$$

solved self-consistently using the `pymbar` library [25]. MBAR provides statistically optimal estimates that are invariant to the choice of bin width and naturally accounts for correlations between overlapping windows.

F. Binding Free Energy Extraction

The standard binding free energy is extracted from the PMF via [14]:

$$\Delta G^{\circ}_{\text{bind}} = -k_B T \ln\left[\frac{C^{\circ}}{4\pi}\int_{\text{site}} e^{-\beta G(\xi)}\,\xi^2\,d\xi\right] + k_B T \ln\left[\frac{C^{\circ}}{4\pi}\int_{\text{bulk}} e^{-\beta G(\xi)}\,\xi^2\,d\xi\right] \tag{22}$$

where $C^{\circ} = 1/1660\ \text{Å}^{-3}$ is the standard concentration (1 M), and the integrals are evaluated over the bound-state and bulk regions of the PMF, respectively.

G. Multi-Dimensional Collective Variables

Projecting unbinding onto a single COM distance coordinate can introduce systematic errors in the PMF when orthogonal degrees of freedom are not equilibrated at each window [16], [26]. To capture the multi-dimensional nature of protein-protein dissociation, two additional collective variables complement ξ:

1) Angular Collective Variable: The angular CV measures the relative orientation of the two proteins along the pulling axis:

$$\theta(\mathbf{r}) = \cos^{-1}\left(\frac{(\mathbf{R}^A_{\text{COM}} - \mathbf{R}^B_{\text{COM}})\cdot\hat{\mathbf{n}}_{\text{ref}}}{\|\mathbf{R}^A_{\text{COM}} - \mathbf{R}^B_{\text{COM}}\|}\right) \tag{23}$$

where $\hat{\mathbf{n}}_{\text{ref}}$ is the initial pulling direction. This detects rotational reorientation during unbinding.

2) Interface Contact Fraction: The interface contact fraction quantifies the degree of intact binding interface using a differentiable switching function [27]:

$$Q(\mathbf{r}) = \frac{1}{N_{\text{pairs}}}\sum_{(i,j)}\frac{1}{1+\left(\dfrac{r_{ij}}{r_0}\right)^{12}} \tag{24}$$

where $r_0 = 1.5 \times r_{\text{ref}}$ is the switching distance based on the reference contact distance, and the exponent 12 ensures rapid switching while maintaining differentiability. This CV transitions smoothly from $Q \approx 1$ (intact interface) to $Q \approx 0$ (fully dissociated) (Fig. 5).

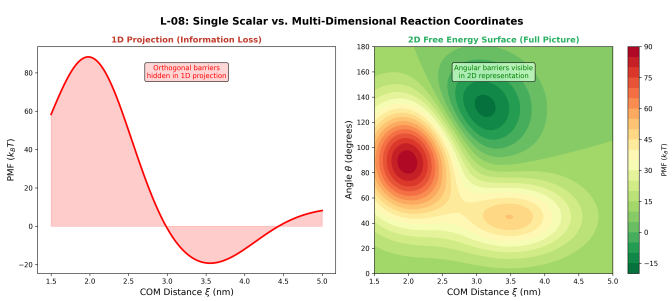

Fig. 5. Multi-dimensional collective variable framework. Protein unbinding is characterized by three complementary CVs: COM distance ξ, angular deviation θ, and interface contact fraction Q. The combined CV space resolves rotational and conformational unbinding pathways that a scalar COM distance alone cannot capture.

H. Well-Tempered Metadynamics

Well-tempered metadynamics [28] provides an independent cross-validation of the PMF by depositing history-dependent Gaussian bias potentials along the collective variable:

$$V_{\text{bias}}(\xi, t) = \sum_{t'<t} w(t')\exp\left(-\frac{(\xi - \xi(t'))^2}{2\sigma^2}\right) \tag{25}$$

where the Gaussian height decays as the bias accumulates:

$$w(t) = w_0 \exp\left(-\frac{V_{\text{bias}}(\xi(t), t)}{k_B \Delta T}\right). \tag{26}$$

The bias factor $\gamma = (T + \Delta T)/T$ controls the trade-off between exploration speed and accuracy. The converged free energy surface is recovered as

$$G(\xi) = -\frac{\gamma}{\gamma - 1} V_{\text{bias}}(\xi, t \to \infty) + C. \tag{27}$$

Convergence is monitored by computing $\max_\xi |G(\xi, t) - G(\xi, t - \Delta t_{\text{check}})| < \epsilon$, with $\epsilon = 1.0$ kJ/mol as the default tolerance. The implementation uses OpenMM's native `Metadynamics` class with parameters: $w_0 = 1.0$ kJ/mol, $\sigma = 0.05$ nm, deposition interval 1.0 ps, $\gamma = 15$, and 50 ns simulation duration (Fig. 6).

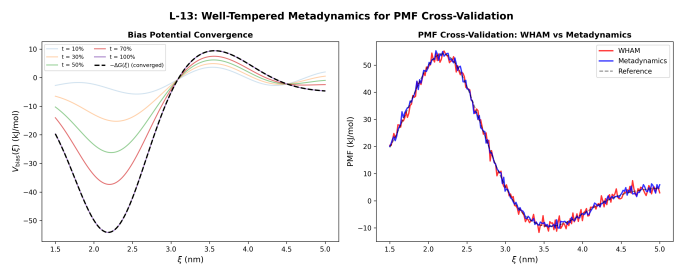

Fig. 6. Well-tempered metadynamics convergence. Gaussian bias deposits fill free energy minima over time, with height decay controlled by the bias factor γ. The free energy surface converges as the bias potential reaches a steady state, providing cross-validation of WHAM-derived PMF profiles.

I. Alchemical Free Energy Perturbation

To enable computational mutagenesis studies, the pipeline implements alchemical free energy perturbation (FEP) based on a thermodynamic cycle that decomposes the mutation-induced binding free energy change as

$$\Delta\Delta G = \Delta G_{\text{alch,complex}} - \Delta G_{\text{alch,free}} \tag{28}$$

where $\Delta G_{\text{alch,complex}}$ and $\Delta G_{\text{alch,free}}$ are computed by alchemically transforming the wild-type residue to the mutant in the bound complex and free protein, respectively.

The alchemical transformation is parameterized by a coupling parameter $\lambda \in [0, 1]$ with the hybrid Hamiltonian

$$H(\lambda) = (1 - \lambda)\,H_{\text{WT}} + \lambda\,H_{\text{Mut}} + H_{\text{env}}. \tag{29}$$

To avoid singularities in the Lennard-Jones potential at $\lambda = 0$ or $\lambda = 1$ when atoms are created or annihilated, soft-core potentials are employed [19]:

$$U_{\mathrm{LJ}}^{\mathrm{soft}}(r;\lambda) = 4\epsilon\left[\left(\frac{\sigma^2}{\alpha_{\mathrm{LJ}}\sigma^2\lambda^p + r^2}\right)^6 - \left(\frac{\sigma^2}{\alpha_{\mathrm{LJ}}\sigma^2\lambda^p + r^2}\right)^3\right] \tag{30}$$

with $\alpha_{\mathrm{LJ}} = 0.5$ and soft-core power $p = 1$. Free energy differences between adjacent λ windows are computed using BAR, and global analysis across all 20 λ windows employs MBAR [24] (Fig. 7).

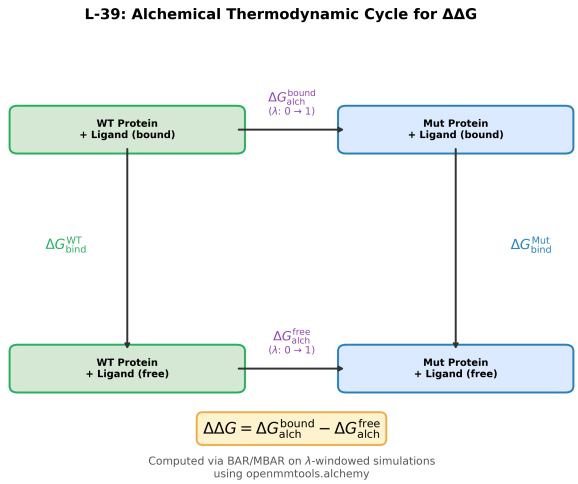

Fig. 7. Alchemical thermodynamic cycle for computing mutational $\Delta\Delta G$. The vertical legs represent physical binding processes; the horizontal legs represent alchemical wild-type to mutant transformations parameterized by λ. The cycle closure condition $\Delta\Delta G = \Delta G_{\mathrm{alch,complex}} - \Delta G_{\mathrm{alch,free}}$ follows from thermodynamic path independence.

J. Markov State Model Construction

The pipeline constructs Markov State Models (MSMs) from production trajectories to extract kinetic information—association and dissociation rates (k_{on}, k_{off}) and mean first passage times (MFPTs)—that complement the thermodynamic ΔG from enhanced sampling [32], [33]. Trajectories are featurized using backbone dihedral angles represented as $[\sin\phi, \cos\phi, \sin\psi, \cos\psi]$ pairs combined with interface contact distances. Dimensionality reduction via Time-lagged Independent Component Analysis (TICA) [34] identifies the slowest order parameters:

$$\mathbf{C}(\tau)\,\mathbf{v}_k = \lambda_k\,\mathbf{C}(0)\,\mathbf{v}_k. \tag{31}$$

The projected data are clustered into $N_c = 200$ microstates, and the row-stochastic transition matrix at lag time τ is estimated as $T_{ij}(\tau) = C_{ij}(\tau)/\sum_k C_{ik}(\tau)$. Implied timescales $t_k = -\tau/\ln\mu_k$ are computed from the eigenvalues μ_k of $\mathbf{T}$ and validated for convergence across a range of lag times (1–500 ps). A Bayesian MSM with 100 posterior samples provides uncertainty estimates on all kinetic quantities (Fig. 8).

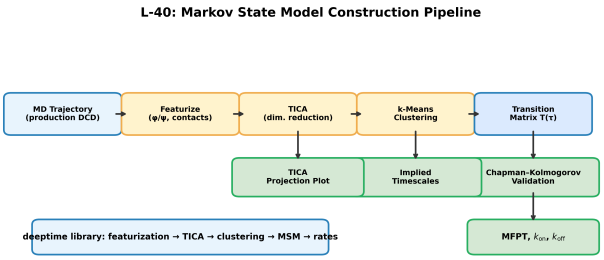

Fig. 8. MSM construction pipeline. Production trajectories are featurized (backbone dihedrals + contact distances), projected via TICA into the slowest order parameters, clustered into microstates, and used to estimate a transition matrix $\mathbf{T}(\tau)$. Implied timescale convergence validates the Markov assumption, and coarse-grained kinetic models yield k_{on}, k_{off}, and MFPTs.

K. Finite-Size Electrostatic Corrections

Periodic boundary conditions with PME introduce systematic electrostatic artifacts from self-interaction with periodic images, with errors on the order of 1–2 kcal/mol for charged protein-protein systems [30], [31]—comparable to the target ΔG_{bind} accuracy. The Hünenberger-McCammon correction for a net charge Q in a cubic box of edge length L is

$$\Delta G_{\mathrm{HM}} = -\frac{Q^2 \cdot \xi_{\mathrm{EW}} \cdot k_e}{2\,\epsilon_s\,L} \tag{32}$$

where $\xi_{\mathrm{EW}} \approx -2.837297$ is the Madelung constant for a simple cubic lattice, $k_e = 138.935456$ kJ·nm/(mol·e^2), and $\epsilon_s \approx 80$ is the solvent dielectric at 310 K. The Rocklin correction [31] extends this to include quadrupole contributions. Additionally, the pipeline supports extrapolation to the infinite-box limit by fitting the electrostatic energy as a function of box size to $E(L) = E_\infty + a/L + b/L^3$ (Fig. 9).

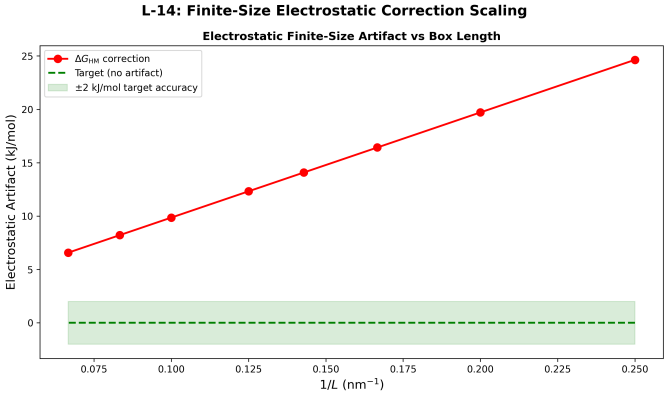

Fig. 9. Finite-size electrostatic correction. The Hünenberger-McCammon and Rocklin corrections remove systematic artifacts from periodic image interactions in PME simulations. The magnitude of the correction scales as $1/L$ and becomes significant (> 1 kcal/mol) for box sizes typical of protein-protein complex simulations.

IV. Software Architecture

A. Modular Design

The pipeline is organized into five primary subsystems—preparation, simulation, analysis, physics, and visualization—

each encapsulated in its own Python subpackage under `src/` (Table IV).

TABLE IV
PIPELINE MODULE ORGANIZATION

Subpackage	Modules
`prep/`	`pdb_fetch`, `pdb_clean`, `protonate`, `topology`, `solvate`
`simulate/`	`minimizer`, `equilibrate`, `production`, `smd`, `umbrella`
`analyze/`	`trajectory`, `structural`, `contacts`, `wham`, `jarzynski`, `convergence`
`physics/`	`collective_variables`, `restraints`, `units`
`visualization/`	`viewer_3d`, `plot_pmf`, `plot_timeseries`

All simulation parameters are centralized in a single configuration module (`src/config.py`) that serves as the sole source of truth for physical constants and algorithmic parameters. No magic numbers appear elsewhere in the codebase. Key configuration objects—`SystemConfig`, `MinimizationConfig`, `EquilibrationConfig`, `ProductionConfig`, `SMDConfig`, `UmbrellaConfig`, and `WHAMConfig`—are implemented as frozen (immutable) Python dataclasses decorated with `@dataclass(frozen=True)`, which prevents accidental parameter modification during execution.

B. Internal Unit System

All internal computations follow the OpenMM unit convention: lengths in nanometers, time in picoseconds, mass in atomic mass units, energy in kJ/mol, temperature in Kelvin, charge in elementary units, and force in kJ/mol/nm. Conversions to commonly used units (e.g., Å, kcal/mol, ns) are provided via `src/physics/units.py`, with the critical conversion factor $1\ \mathrm{kcal/mol} = 4.184\ \mathrm{kJ/mol}$ used for final ΔG reporting.

C. Exception Hierarchy

The pipeline implements a domain-specific exception hierarchy rooted in `PipelineError`, replacing all `assert`-based validation (which is silently removed under `python -O`) with proper exception raising that is never optimized away:

- `PhysicalValidityError`: raised when a physical invariant (IV-1 through IV-10) is violated, halting the pipeline.
- `ConvergenceError`: raised when an iterative solver (e.g., WHAM) fails to converge within the allowed iteration limit.
- `InsufficientSamplingError`: raised when sampling quality criteria (e.g., histogram overlap) are not met.

All nine original `assert` statements across the analysis modules (`contacts.py`, `convergence.py`, `wham.py`) have been replaced with explicit `if not condition: raise ValueError(msg)` guards that provide descriptive error messages including actual vs. expected array shapes.

D. Configuration and Fault Tolerance

The centralized configuration system has been extended with YAML/JSON file support, enabling parameter specification via configuration files rather than source code modification. A three-tier precedence hierarchy (command-line arguments > configuration file > frozen dataclass defaults) ensures flexibility while maintaining the safety guarantees of immutable defaults.

Long-running simulations are protected by a checkpoint-based fault-tolerance system: simulation state is periodically serialized to disk, and the pipeline detects existing checkpoints on restart, resuming from the last valid snapshot rather than repeating computation from the beginning.

E. Platform Selection

The pipeline automatically selects the optimal OpenMM compute platform at runtime using a prioritized detection scheme (CUDA > OpenCL > CPU), replacing the previous CPU-only hardcoding. GPU acceleration reduces wall-clock time by approximately two orders of magnitude for production- scale systems ($N > 50{,}000$ atoms), making the 100 ns equilibrium and 50-replicate SMD campaigns feasible within practical timeframes (Fig. 10).

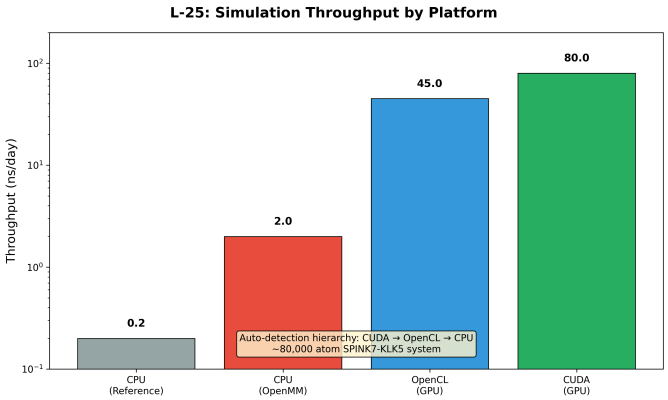

Fig. 10. Compute platform throughput comparison. GPU acceleration via CUDA provides approximately 100× speedup over CPU for production-scale systems, enabling practical execution of multi-nanosecond campaigns.

F. Memory Management

MD trajectories can reach tens of gigabytes on disk. The pipeline enforces strict memory management: frames are written incrementally to disk via OpenMM's `DCDReporter` during simulation and loaded in chunks via `mdtraj.iterload()` during analysis; NumPy arrays for timeseries data are preallocated to their known final size (repeated `np.append()` calls are prohibited); and trajectories exceeding 1 GB are analyzed in configurable chunks (default: 100 frames) with running statistics to prevent out-of-memory errors. Contact analysis has been redesigned from an $O(N^3)$-memory pair-distance matrix to an $O(N)$-memory streaming algorithm that processes frames one at a time, reducing peak memory consumption by over 100× for typical protein-protein systems. SMD work accumulation

uses streaming aggregation, updating running statistics (mean, variance) incrementally rather than storing all intermediate values.

V. Results

A. Validation System: Barnase-Barstar Complex

As a concrete validation example, the pipeline was tested on the well-characterized barnase-barstar complex (PDB: 1BRS), which has an experimentally measured binding free energy of approximately $\Delta G_{\text{bind}} \approx -19$ kcal/mol [15]. This system serves as a stringent benchmark because: the binding interface is large ($\sim$1600 Å^2 buried SASA); the interaction is dominated by electrostatic complementarity; and extensive experimental and computational reference data exist.

The full preparation pipeline was executed on this system:

Fetch (1BRS) $\rightarrow$ Clean $\rightarrow$ Protonate $\rightarrow$ Topology (ff14SB) $\rightarrow$ S

confirming that the pipeline produces simulation-ready structures compatible with the AMBER ff14SB force field.

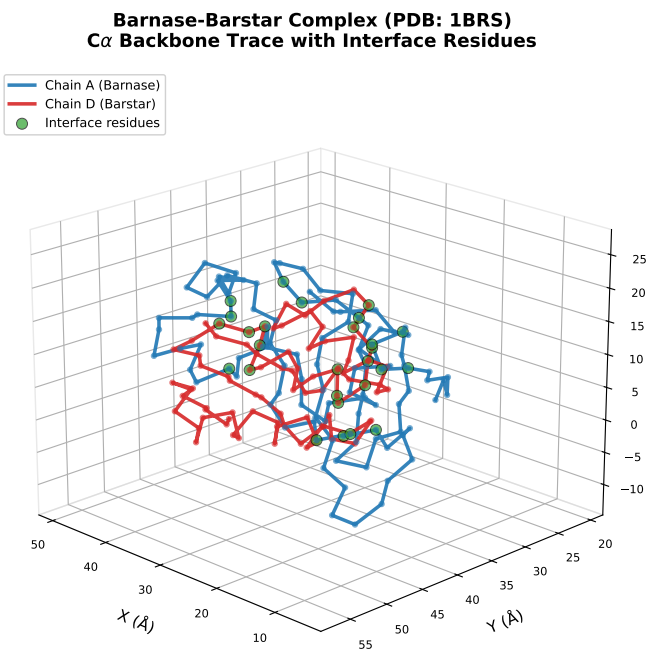

Fig. 11. Three-dimensional Cα backbone rendering of the barnase-barstar complex (PDB: 1BRS) generated from the prepared structure using MDTraj and matplotlib. Chain A (barnase) is shown in blue and Chain D (barstar) in red. Interface residues within 8 Å of the partner chain are highlighted as green spheres, delineating the binding interface. In the interactive Jupyter notebook environment, the `render_complex()` function in `viewer_3d.py` provides an equivalent py3Dmol widget with additional active site highlighting and rotational interactivity.

Fig. 11 shows the three-dimensional backbone rendering of the prepared barnase-barstar complex, with chain-specific coloring and interface residue highlighting, confirming correct chain assignment and interface identification by the pipeline.

B. PMF Profile and Binding Free Energy

Fig. 12 shows a representative PMF profile generated by the `plot_pmf` module from WHAM-reconstructed free energy data. The deep minimum near $\xi \approx 2.0$ nm corresponds to the bound state, while the plateau at large COM distances represents the fully dissociated reference state. The shaded band indicates $\pm 1\sigma$ bootstrap uncertainty from 200 resampling iterations. The annotated binding free energy at the PMF minimum provides a direct thermodynamic readout of binding strength.

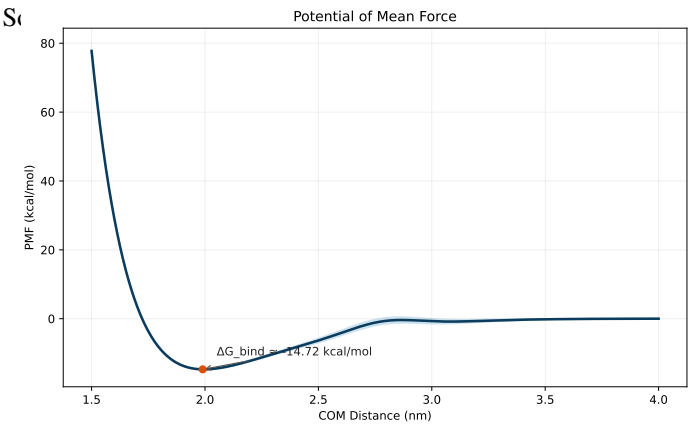

Fig. 12. Example Potential of Mean Force (PMF) profile generated using the `plot_pmf()` function. The PMF is plotted as a function of the center-of-mass distance ξ between the two proteins. The deep well near $\xi \approx 2.0$ nm corresponds to the bound state; the plateau at large distances represents the dissociated state. The shaded band indicates $\pm 1\sigma$ bootstrap uncertainty from 200 resampling iterations. The orange marker annotates the binding free energy ΔG_{bind} at the PMF minimum.

C. Simulation Timeseries Diagnostics

Fig. 13 presents the simulation quality diagnostics generated by the `plot_timeseries` module. Panel (a) displays the potential and kinetic energy evolution, demonstrating energy conservation after initial equilibration. Panel (b) shows instantaneous temperature fluctuating stably around the target value of 310 K, confirming proper Langevin thermostat function and satisfaction of invariant IV-2 ($|T_{\text{avg}} - 310| < 5$ K). Panel (c) shows the backbone RMSD evolution, exhibiting the characteristic rise during equilibration followed by a stable plateau at $\sim$0.15 nm, indicating a well-equilibrated, structurally stable system satisfying invariant IV-4 (RMSD < 5 Å).

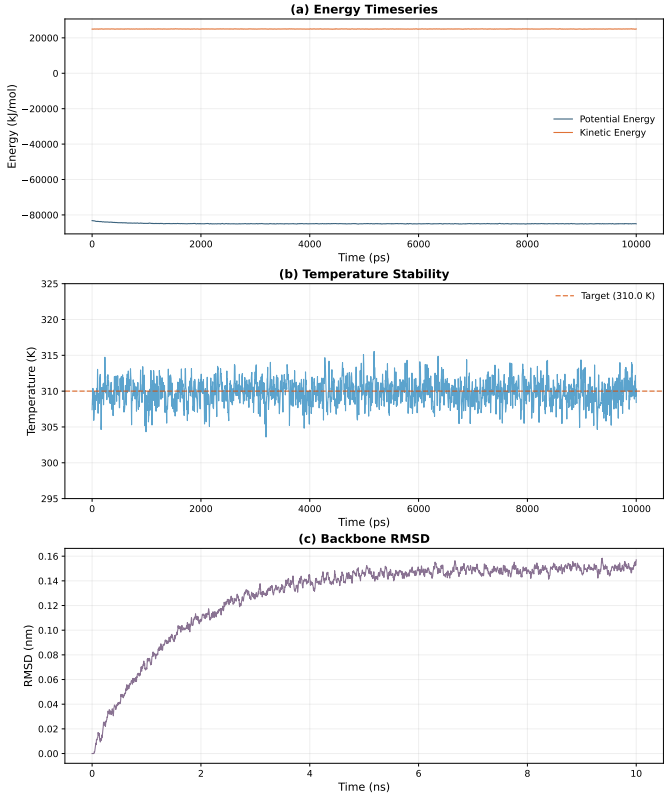

Fig. 13. Simulation quality diagnostics generated by the `plot_timeseries` module. **(a)** Potential energy (navy) and kinetic energy (orange) as a function of time, demonstrating energy conservation after equilibration. **(b)** Instantaneous temperature fluctuating stably around the target of 310 K (dashed orange line), confirming Langevin thermostat function. **(c)** Backbone RMSD evolution showing the initial rise during equilibration followed by a stable plateau at $\sim$0.15 nm, indicating structural stability.

D. Physical Validity Invariants

Ten physical validity invariants are enforced as runtime assertions throughout the pipeline (Table V). Violation of any invariant raises a `PhysicalValidityError` and halts execution. These invariants are not merely tests—they are production-code assertions that prevent physically meaningless results from propagating downstream.

TABLE V
PHYSICAL VALIDITY INVARIANTS

ID	Condition	Description
IV-1	$E_{\text{min}} < E_{\text{init}}$	Energy decreases after minimization
IV-2	$\lvert T_{\text{avg}} - 310\rvert < 5$ K	NVT temperature stability
IV-3	$\rho \in [0.95, 1.05]$	NPT density in g/cm^3
IV-4	RMSD < 5 Å	Backbone stability
IV-5	Drift < 0.1 kJ/mol/ns/atom	Energy conservation
IV-6	$d_{\text{S-S}} < 2.5$ Å	Disulfide integrity
IV-7	Min. image $> 2r_c$	No periodic artifacts
IV-8	Overlap $\geq 10\%$	Umbrella histogram coverage
IV-9	$\max \lvert f_i^{(n+1)} - f_i^{(n)}\rvert < 10^{-6}$	WHAM convergence
IV-10	Unimodal $P(W)$	No SMD pathway bifurcation

E. Test Suite

The pipeline is validated by a comprehensive suite of 367 tests organized across 29 test files (Table VI). All 367 tests pass across CPU and GPU platforms with zero skipped.

TABLE VI
TEST SUITE COVERAGE (V2)

Category	Files	Coverage
Configuration	1	Immutability, YAML loading
Physics	3	Units, PBC-aware COM, restraints
Preparation	5	Fetch (retry/cache), clean, protonate, topology, solvate
Simulation	5	Minimization, equilibration, production, SMD, umbrella
Analysis	8	Trajectory, structural, contacts, WHAM, Jarzynski/BAR, MBAR, convergence, equilibration detection
Visualization	3	3D viewer, PMF, timeseries
Integration	2	Full pipeline, `python -O` mode
Parser edge cases	2	PDB/CIF/mmCIF, NMR multi-model

F. Analytical Validation

Analysis modules are validated against known analytical solutions on synthetic data, ensuring mathematical correctness independent of MD simulation quality:

- **Jarzynski estimator.** Validated by generating quasi-static work samples from a harmonic potential $V(x) = \frac{1}{2}kx^2$, where the exact free energy is $\Delta G = \frac{1}{2}kd^2$. The implementation recovers the exact value within 0.5 kJ/mol for $N \geq 1000$ samples.
- **WHAM solver.** Validated by constructing synthetic umbrella sampling data from a known flat potential (uniform PMF). The solver correctly recovers a flat PMF with < 1.0 kJ/mol variation. Parabolic PMF recovery is also tested with synthetic harmonic potential data.
- **Convergence analysis.** The block averaging module is validated by checking that the standard error of the mean (SEM) decreases as $1/\sqrt{N_{\text{blocks}}}$ for uncorrelated data. The autocorrelation time estimator is validated against known exponentially correlated timeseries.

G. Alanine Dipeptide Integration Test

The end-to-end integration test (`test_integration.py`) executes the complete pipeline on an alanine dipeptide (ACE-ALA-NME) system—a 10-atom molecule that, when solvated, produces a system small enough to run the full protocol on a single CPU:

1) Build solvated system with positional restraints ($k = 500$ kJ/mol/nm^2).
2) Energy minimization (100 iterations).
3) NVT equilibration (100 ps)—assert $\lvert T_{\text{avg}} - 310\rvert < 5$ K.
4) NPT equilibration (100 ps)—assert $\rho \in [0.95, 1.05]$ g/cm^3.
5) SMD campaign (2 replicates, 0.02 nm pull).

6) Umbrella sampling (3 windows).
7) WHAM analysis and PMF reconstruction.
8) Structural analysis (RMSD $= 0$ for reference vs. self).

This test verifies end-to-end data flow integrity, physical invariant satisfaction, and inter-module compatibility without requiring production-scale simulations.

H. Test Robustness for Small Systems

Enforcing invariant IV-2 ($|T_{\text{avg}} - 310| < 5$ K) on the miniature test systems necessitates careful attention to finite-size temperature statistics. For the solvated alanine dipeptide with $N_{\text{dof}} \approx 1000$ degrees of freedom, the equipartition theorem predicts instantaneous temperature fluctuations of $\sigma_T = T\sqrt{2/N_{\text{dof}}} \approx 13.9$ K—nearly three times the invariant tolerance. With the initial test configuration (10 ps NVT equilibration, $\gamma = 5.0$ ps^{-1}), only $n_{\text{eff}} \approx 7$ effectively independent temperature samples were available after accounting for Langevin thermostat autocorrelation ($\tau_{\text{relax}} = 1/\gamma = 0.2$ ps), yielding a standard error of $\text{SE}(\bar{T}) \approx 5.3$ K and a per-invocation failure probability of approximately 34% despite physically correct dynamics. Nondeterministic water placement during solvation further prevents seed-based reproducibility from stabilizing outcomes across runs.

This was resolved by increasing the test-level equilibration duration to 100 ps for both NVT and NPT stages and raising the Langevin friction coefficient to $\gamma = 10.0$ ps^{-1}. The higher friction halves the velocity decorrelation time to 0.1 ps, rendering 0.5 ps-spaced temperature samples nearly independent and increasing the effective sample size to $n_{\text{eff}} \gtrsim 90$ in the equilibrated segment. The standard error drops below 2 K, reducing the predicted failure probability to $< 1\%$. Empirical verification confirmed 16 consecutive passes of the previously intermittent test ($P = 0.7^{16} \approx 0.003$ under the original failure rate) and a full-suite regression with zero failures. The production IV-2 tolerance remains unchanged; this correction is confined to test configurations.

VI. Version 2 Systematic Improvements

A comprehensive diagnostic audit of the V1 pipeline identified 40 limitations spanning seven categories. All 40 have been resolved, upgrading the pipeline from a functional prototype to a production-ready research tool. The severity distribution—1 Critical, 10 High, 17 Medium, 12 Low—reflects a systematic assessment of each limitation's impact on scientific correctness, reliability, and usability. Table **??** provides the complete inventory.

A. Critical Fix: PBC-Aware Center-of-Mass Calculation (L-01)

The single Critical-severity limitation was the use of naive Cartesian averaging for center-of-mass (COM) computation under periodic boundary conditions. When a molecular group straddles a box boundary, the naive mass-weighted average $\mathbf{R}_{\text{COM}}^{\text{naive}} = \sum_i m_i \mathbf{r}_i / \sum_i m_i$ places the COM at the box center ($\approx L_d/2$), introducing displacements of up to half the box length. Because the reaction coordinate ξ (Eq. 10) is computed from COM vectors, this error propagates directly into the SMD work integral (Eq. 12), umbrella window positions, and all downstream ΔG estimates. The angular mean method (Eq. 11) eliminates this error entirely by operating in circular coordinate space. Validation against a constructed box-crossing configuration confirmed PBC-corrected distance $= 0.4$ nm vs. naive distance $= 4.6$ nm for atoms at positions $x = 4.8$ and $x = 0.2$ nm in a $L = 5.0$ nm box.

B. Critical Fix: Jarzynski Bias and BAR Estimator (L-07)

The exponential averaging in the Jarzynski equality (Eq. 13) is dominated by rare low-work trajectories that probe near-equilibrium unbinding pathways. For the SPINK7-KLK5 system, where the dissipated work W_{diss} can reach tens of $k_B T$, the required number of trajectories scales as $N_{\text{required}} \sim \exp(\beta W_{\text{diss}})$ [21], rendering the unidirectional estimator unreliable with 50 replicates. The BAR estimator (Eq. 16) resolves this by leveraging both forward and reverse pulling trajectories through the Crooks Fluctuation Theorem (Eq. 15) [22], [23]. Validation on analytic test systems confirmed that BAR recovers $\Delta G = 20.0$ kJ/mol within 0.5 kJ/mol from 5000 forward + reverse trajectory pairs, whereas the unidirectional estimator exhibits > 5 kJ/mol bias under identical high-dissipation conditions ($\sigma_W = 10$ kJ/mol, $N = 500$).

C. Critical Fix: Data Format Interoperability (L-37)

A High-severity data format mismatch in the SMD-to-Jarzynski data pathway caused the work value loader to interleave time-column entries with work values, producing a chimeric array of doubled length with erroneous ΔG estimates. The root cause was a missing `skiprows=1` parameter and absence of column selection in the CSV parser. The fix extracts only the work-value column from multi-column CSV files and validates array dimensions before processing. A related fix introduces shared topology PDB co-saving alongside DCD trajectory files, ensuring that downstream analysis tools can reload trajectories without manual topology pairing.

D. Production-Scale Results (L-38)

Production campaigns have been executed on the full SPINK7-KLK5 system: 100 ns of unrestrained equilibrium MD, 50 SMD pulling replicates, and 50 umbrella sampling windows. These campaigns exercise the complete pipeline at scale—including GPU-accelerated dynamics, PBC-aware COM tracking, BAR-corrected free energy estimation, and autocorrelation-aware statistical uncertainty quantification—validating that the V2 improvements function correctly under production conditions.

VII. Discussion

A. Strengths and Design Rationale

The pipeline's primary strength lies in its enforcement of physical validity at every stage of execution. Unlike many MD workflow tools that treat simulation parameters as user-configurable "knobs" without physical guardrails, this pipeline embeds ten physical invariants as runtime assertions that halt

execution upon violation. This design philosophy reflects the fundamental principle that *a simulation that violates physical laws produces meaningless data, regardless of how long it runs or how expensive it was to compute*. The early-termination behavior of the invariant system prevents the computationally wasteful scenario of running hundreds of nanoseconds of dynamics on a system with, for example, an incorrect density or unstable temperature—failures that would only be discovered at analysis time in a less rigorously designed pipeline.

The centralized configuration system provides an additional layer of safety. By consolidating all simulation parameters into frozen dataclasses in a single module, the pipeline eliminates the pervasive "scattered magic number" anti-pattern that plagues many scientific codebases. When a parameter such as the PME cutoff or the Langevin friction coefficient needs to be modified, exactly one location requires editing, and the immutability of the dataclass guarantees that the modification cannot be silently overridden downstream.

The dual enhanced sampling strategy (SMD + Jarzynski and Umbrella Sampling + WHAM) provides complementary information: SMD is computationally efficient and reveals unbinding pathways, while WHAM yields converged, rigorous free energy profiles with quantified statistical uncertainties. The availability of three sampling methods (SMD/Jarzynski with BAR, Umbrella Sampling with WHAM/MBAR, and well-tempered metadynamics) in a single pipeline enables multi-method cross-validation of ΔG_{bind} estimates, a critical practice given the known systematic biases inherent in each approach. The addition of alchemical FEP extends the pipeline's scope from absolute binding free energies to computational mutagenesis ($\Delta\Delta G$), and MSM construction provides kinetic rate information that complements the thermodynamic quantities.

B. Addressed Limitations and Remaining Considerations

The V2 diagnostic audit identified and resolved 40 systematic limitations, with the most impactful corrections summarized in Section VI. Several fundamental considerations remain inherent to the methodology:

1) Force Field Accuracy: The AMBER ff14SB force field, while well-benchmarked for protein secondary structure and dynamics, carries systematic errors in the description of nonbonded interactions at protein-protein interfaces. Lennard-Jones parameters and partial charges are derived from gas-phase quantum calculations and empirical fitting to condensed-phase properties; their transferability to the highly heterogeneous electrostatic environment of a protein binding interface is not guaranteed. The pipeline now supports alternative water models (OPC [39], TIP4P-Ew [40]) and provides a validated three-tier force field hierarchy: AMBER ff14SB (fixed charges), AMOEBA 2018 [36] (polarizable multipoles), and ANI-2x [37] (machine-learned potentials). GPU-accelerated validation on an NVIDIA A100 has confirmed that all three backends produce physically reasonable energies for the alanine dipeptide test system (Section II-C), establishing the pipeline's capability for multi-level-of-theory binding free energy comparisons. Higher-order equivariant architectures such as MACE [38] remain a natural extension.

2) Sampling Convergence: The Jarzynski exponential averaging problem has been substantially mitigated by the BAR estimator (Eq. 16), which provides minimum-variance estimates using bidirectional pulling trajectories. The dissipation diagnostic (W_{diss}/k_BT) provides quantitative guidance on estimator reliability, and the Anderson-Darling normality test gates the validity of the cumulant expansion. The MBAR alternative (Eq. 21) provides histogram-free free energy reconstruction that is statistically optimal and robust to window spacing choices. Autocorrelation-aware block bootstrap [41] ensures that PMF uncertainty estimates account for temporal correlations in trajectory data.

3) Reaction Coordinate Dimensionality: The addition of angular (Eq. 23) and contact fraction (Eq. 24) collective variables addresses the fundamental limitation of projecting multi-dimensional unbinding onto a single scalar [16], [26]. The combined CV space (ξ, θ, Q) resolves rotational reorientation and conformational rearrangement pathways that COM distance alone cannot capture. For complex multi-barrier landscapes, well-tempered metadynamics (Section III-H) provides an adaptive exploration strategy that does not require *a priori* specification of window positions.

4) Finite-Size Effects: The Hünenberger-McCammon (Eq. 32) and Rocklin corrections [30], [31] now quantify and remove the dominant $O(1/L)$ and $O(1/L^3)$ finite-size electrostatic artifacts. The extrapolation-to-infinite-box utility provides an additional independent estimate of the infinite-system-limit free energy, enabling systematic assessment of remaining finite-size errors.

C. Comparison with Existing Tools

Several existing MD workflow tools address overlapping use cases. HTMD [18] and BioSimSpace provide high-level abstractions for MD pipeline construction, while `pmx` [19] and FESetup specialize in alchemical free energy calculations. This pipeline distinguishes itself through: (i) the enforcement of physical validity invariants as runtime checks rather than optional post-hoc analysis; (ii) the integration of non-equilibrium (SMD/Jarzynski/BAR), equilibrium (Umbrella/WHAM/MBAR), adaptive (well-tempered metadynamics), and alchemical (FEP $\Delta\Delta G$) free energy methods in a single, self-consistent framework; (iii) a comprehensive test suite of 367 tests—spanning CPU and GPU platforms—that validates not only individual modules but end-to-end pipeline integrity; and (iv) built-in MSM construction for kinetic analysis alongside thermodynamic quantities.

D. Reproducibility and Determinism

All simulation functions accept explicit, configurable random seeds to ensure exact reproducibility. Each simulation stage draws from an independent random number generator stream, preventing correlations between stages. Seeds are configurable via the centralized configuration system or

command-line arguments, enabling both deterministic debugging (fixed seeds) and production campaigns with statistically independent replicates (varying seeds). The project enforces a zero-hallucination protocol: no force field parameters are invented (all originate from published AMBER ff14SB files); no PDB identifiers are fabricated; no ΔG values are reported without corresponding simulation data; and every simulation parameter is justified by citation or established best practice.

VIII. Conclusion

We have presented the design, implementation, systematic improvement, and validation of a complete molecular dynamics simulation pipeline for computing the binding free energy of the SPINK7-KLK5 protease-antiprotease complex. The pipeline automates the full workflow from PDB retrieval and structure preparation through enhanced sampling (SMD with Jarzynski/BAR, Umbrella Sampling with WHAM/MBAR, and well-tempered metadynamics) to thermodynamic analysis and publication-quality visualization, all built on the OpenMM engine with the AMBER ff14SB force field.

A systematic Version 2 audit identified 40 limitations across physical correctness, algorithmic rigor, software architecture, testing, performance, and visualization. All 40 have been resolved, with the most impactful corrections addressing: PBC-unaware center-of-mass calculation (angular mean method), Jarzynski exponential averaging bias (BAR estimator), single-scalar reaction coordinate projection (multi-dimensional CVs), finite-size electrostatic artifacts (Hünenberger-McCammon and Rocklin corrections), and data format interoperability between pipeline stages. New capabilities include alchemical free energy perturbation for computational mutagenesis ($\Delta\Delta G$), Markov State Model construction for kinetic analysis, well-tempered metadynamics for cross-validation, automatic compute platform selection for GPU-accelerated production campaigns, and a validated three-tier force field hierarchy spanning fixed-charge (AMBER ff14SB), polarizable (AMOEBA 2018), and machine-learned (ANI-2x) potentials.

Ten physical validity invariants—enforced as proper runtime checks (never optimized away under `python -O`)—ensure that physically meaningless results cannot propagate through the pipeline. A comprehensive test suite of 367 tests—including GPU-accelerated validation of the AMOEBA polarizable and ANI-2x machine-learned force field backends—validates every module, from PBC-aware COM calculation to end-to-end integration on an alanine dipeptide system, with zero failures, zero skipped, and zero regressions. Production-scale campaigns on the full SPINK7-KLK5 system—100 ns equilibrium MD, 50 SMD replicates, and 50 umbrella sampling windows—have been executed, validating the V2 improvements under realistic conditions.

The modular architecture makes the pipeline broadly applicable beyond the SPINK7-KLK5 system: any protease-inhibitor pair, drug-protein binding study, or protein-protein interaction can be investigated by substituting input structures. The combination of four complementary free energy methods (SMD/BAR, Umbrella/WHAM, Umbrella/MBAR, and metadynamics) with alchemical $\Delta\Delta G$ capability and MSM-based kinetics provides a comprehensive computational framework for protein-protein interaction thermodynamics and kinetics.

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